

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**Product Identifier**

**Product Description:** 2-Isopropoxyethanol  
**Cat No. :** S33753  
**Synonyms** Isopropyl Cellosolve™ 4; Ethylene glycol monoisopropylether  
**CAS No** 109-59-1  
**Molecular Formula** C5 H12 O2

**Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**Details of the supplier of the safety data sheet**

|                                                                                                                                                                                               |                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Importer</b><br>Fisher Scientific Korea<br>D5,D6, Incheon Airport Logistics Complex<br>150, Gonghangdong-Ro 296 Beon-Gil<br>Jung-Gu, Incheon<br>Tel: +82-1661-9555<br>Fax: +82-2-2023-0603 | <b>Supplier</b><br>Thermo Fisher Scientific Chemicals, Inc.<br>30 Bond Street<br>Ward Hill, MA 01835-8099 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|

**E-mail address** Chem.KR@thermofisher.com

**Emergency Telephone Number**

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970  
 CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

## SECTION 2: HAZARDS IDENTIFICATION

**Classification of the substance or mixture**
**Physical hazards**

Flammable liquids Category 3

**Health hazards**

|                                    |            |
|------------------------------------|------------|
| Acute Dermal Toxicity              | Category 4 |
| Acute Inhalation Toxicity - Vapors | Category 4 |
| Serious Eye Damage/Eye Irritation  | Category 2 |

**Environmental hazards**

Based on available data, the classification criteria are not met

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## Label Elements



### Signal Word

### Warning

#### Hazard Statements

H226 - Flammable liquid and vapor

H319 - Causes serious eye irritation

H312 + H332 - Harmful in contact with skin or if inhaled

#### Precautionary Statements

##### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P233 - Keep container tightly closed

P240 - Ground and bond container and receiving equipment

P241 - Use explosion-proof electrical/ ventilating/ lighting equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P271 - Use only outdoors or in a well-ventilated area

P264 - Wash hands and face thoroughly after handling

##### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P312 - Call a POISON CENTER or doctor if you feel unwell

P321 - Specific treatment (see supplemental first aid instructions on this label)

P362 + P364 - Take off contaminated clothing and wash it before reuse

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P337 + P313 - If eye irritation persists: Get medical advice/attention

##### Storage

P403 + P235 - Store in a well-ventilated place. Keep cool

##### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## NFPA

Health  
2

Flammability  
2

Instability  
0

Physical hazards  
N/A

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component | Common Name | CAS No | Index No | Weight % |
|-----------|-------------|--------|----------|----------|
|-----------|-------------|--------|----------|----------|

ALFAAS33753

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|                   |                                                                    |          |          |          |
|-------------------|--------------------------------------------------------------------|----------|----------|----------|
| Isopropoxyethanol | Isopropyl<br>Cellosolve4; Ethylene<br>glycol<br>monoisopropylether | 109-59-1 | KE-23948 | 99 - 100 |
|-------------------|--------------------------------------------------------------------|----------|----------|----------|

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### **Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### **Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention.

#### **Ingestion**

Clean mouth with water. Get medical attention.

#### **Inhalation**

Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Get medical attention.

#### **Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

### Indication of any immediate medical attention and special treatment needed

#### **Notes to Physician**

Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

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## Personal Precautions, Protective Equipment and Emergency Procedures

Remove all sources of ignition. Take precautionary measures against static discharges.

## Environmental precautions

See Section 12 for additional Ecological Information.

## Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Avoid contact with skin and eyes. Do not breathe mist/vapors/spray. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharges.

### Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component         | CAS No   | Korea               | ACGIH TLV           | OSHA PEL                                                      |
|-------------------|----------|---------------------|---------------------|---------------------------------------------------------------|
| Isopropoxyethanol | 109-59-1 | TWA: 25 ppm<br>Skin | TWA: 25 ppm<br>Skin | (Vacated) TWA: 25 ppm<br>(Vacated) TWA: 105 mg/m <sup>3</sup> |

| Component         | CAS No   | European Union | The United Kingdom | Germany                                                                                                                                                                                                                                                                      |
|-------------------|----------|----------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Isopropoxyethanol | 109-59-1 | Not listed     | Not listed         | TWA: 10 ppm (8 Stunden).<br>AGW - exposure factor 2<br>TWA: 44 mg/m <sup>3</sup> (8 Stunden).<br>AGW - exposure factor 2<br>TWA: 10 ppm (8 Stunden).<br>MAK<br>TWA: 43 mg/m <sup>3</sup> (8 Stunden).<br>MAK<br>Höhepunkt: 20 ppm<br>Höhepunkt: 86 mg/m <sup>3</sup><br>Haut |

### ACGIH - Biological Exposure Indices

| Component         | CAS No   | ACGIH - Biological Exposure Indices |
|-------------------|----------|-------------------------------------|
| Isopropoxyethanol | 109-59-1 | Not listed                          |

### Exposure Controls

#### Engineering Measures

Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

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Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

## Personal protective equipment

|                                 |                                                                          |
|---------------------------------|--------------------------------------------------------------------------|
| <b>Eye Protection</b>           | Goggles                                                                  |
| <b>Hand Protection</b>          | Protective gloves                                                        |
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                      |                                                                                                                                                                                                                                                          |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal protective equipment</b> | Use only those certified by the Korea Occupational Safety and Health Administration.                                                                                                                                                                     |
| <b>Respiratory Protection</b>        | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators                                                                                                                                          |
| <b>Recommended Filter type:</b>      | Organic gases and vapours filter Type A Brown conforming to EN14387<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly<br>When RPE is used a face piece Fit Test should be conducted |

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice

Environmental exposure controls No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                                 |                                     |                                          |
|-------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Appearance (Physical State, Color, etc.)</b> | Light yellow Liquid                 |                                          |
| <b>Odor</b>                                     | Odorless                            |                                          |
| <b>Odor Threshold</b>                           | No data available                   |                                          |
| <b>pH</b>                                       | 7                                   |                                          |
| <b>Melting Point/Range</b>                      | -60 °C / -76 °F                     |                                          |
| <b>Softening Point</b>                          | No data available                   |                                          |
| <b>Boiling Point/Range</b>                      | 142 - 144 °C / 287.6 - 291.2 °F     |                                          |
| <b>Flash Point</b>                              | 43 °C / 109.4 °F                    | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                         | No data available                   |                                          |
| <b>Flammability (solid,gas)</b>                 | Not applicable                      | Liquid                                   |
| <b>Explosion Limits</b>                         | <b>Lower</b> 1.6<br><b>Upper</b> 13 |                                          |
| <b>Vapor Pressure</b>                           | No data available                   |                                          |
| <b>Vapor Density</b>                            | No data available                   | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>               | 0.904                               |                                          |
| <b>Bulk Density</b>                             | Not applicable                      | Liquid                                   |
| <b>Water Solubility</b>                         | Soluble                             |                                          |

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Solubility in other solvents No information available

## Partition Coefficient (n-octanol/water)

| Component         | CAS No   | log Pow |
|-------------------|----------|---------|
| Isopropoxyethanol | 109-59-1 | 0.43    |

Autoignition Temperature 345 °C / 653 °F  
Decomposition Temperature No data available  
Viscosity No data available  
Explosive Properties explosive air/vapour mixtures possible  
Oxidizing Properties No information available

Molecular Formula C5 H12 O2  
Molecular Weight 104.15

## SECTION 10: STABILITY AND REACTIVITY

Reactivity None known, based on information available

Chemical Stability Stable under recommended storage conditions.

### Possibility of Hazardous Reactions

Hazardous Polymerization Hazardous polymerization does not occur.  
Hazardous Reactions No information available.

Conditions to Avoid Keep away from open flames, hot surfaces and sources of ignition. Incompatible products.

Incompatible Materials Strong oxidizing agents.

Hazardous Decomposition Products Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### Information on expected route of exposure

Inhalation Avoid breathing vapors or mists. Harmful by inhalation.  
Ingestion May be harmful if swallowed.  
Eyes Avoid contact with eyes. Irritating to eyes. Vapor may cause irritation.  
Skin Avoid contact with skin. May cause irritation. Prolonged skin contact may defat the skin and produce dermatitis. Harmful in contact with skin.

Information on Health Hazards .

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**(a) acute toxicity;**

Oral

Based on available data, the classification criteria are not met

Dermal

Category 4

Inhalation

Category 4

| Component         | CAS No   | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation             |
|-------------------|----------|---------------------------|------------------------------|-----------------------------|
| Isopropoxyethanol | 109-59-1 | LD50 = 5600 mg/kg ( Rat ) | LD50 = 1440 mg/kg ( Rabbit ) | LC50 > 3500 ppm ( Rat ) 4 h |

**(b) skin corrosion/irritation;**

No data available

**(c) serious eye damage/irritation;**

Category 2

**(d) respiratory or skin sensitization;**

Respiratory

No data available

Skin

No data available

| Component         | CAS No   | Test method       | Test species      | Study result      |
|-------------------|----------|-------------------|-------------------|-------------------|
| Isopropoxyethanol | 109-59-1 | No data available | No data available | No data available |

**(e) germ cell mutagenicity;**

No data available

| Component         | CAS No   | Test method       | Test species      | Study result      |
|-------------------|----------|-------------------|-------------------|-------------------|
| Isopropoxyethanol | 109-59-1 | No data available | No data available | No data available |

**(f) carcinogenicity;**

No data available

| Component         | CAS No   | Test method       | Test species / Duration | Study result      |
|-------------------|----------|-------------------|-------------------------|-------------------|
| Isopropoxyethanol | 109-59-1 | No data available | No data available       | No data available |

There are no known carcinogenic chemicals in this product

| Component         | CAS No   | IARC       | NTP        | ACGIH      | OSHA       | UK         |
|-------------------|----------|------------|------------|------------|------------|------------|
| Isopropoxyethanol | 109-59-1 | Not listed | Not listed | Not listed | Not listed | Not listed |

**(g) reproductive toxicity;**

No data available

| Component         | CAS No   | Test method       | Test species / Duration | Study result      |
|-------------------|----------|-------------------|-------------------------|-------------------|
| Isopropoxyethanol | 109-59-1 | No data available | No data available       | No data available |

**(h) STOT-single exposure;**

No data available

**(i) STOT-repeated exposure;**

No data available

Target Organs

No information available.

**(j) aspiration hazard;**

No data available

**Other Adverse Effects**

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

| Component         | CAS No   | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|-------------------|----------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Isopropoxyethanol | 109-59-1 | Not applicable                           | Not applicable                                   | Not applicable                          |

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## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

| Component         | CAS No   | Freshwater Fish                                           | Water Flea                             | Freshwater Algae  | Microtox          |
|-------------------|----------|-----------------------------------------------------------|----------------------------------------|-------------------|-------------------|
| Isopropoxyethanol | 109-59-1 | LC50: > 100 mg/L,<br>96h semi-static<br>(Oryzias latipes) | EC50: = 3610 mg/L,<br>48h (water flea) | No data available | No data available |

### Persistence and degradability

#### Persistence

Soluble in water, Persistence is unlikely, based on information available.

### Bioaccumulative potential

Bioaccumulation is unlikely

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| Isopropoxyethanol | 0.43    | No data available             |

### Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

### Ozone Depletion Potential

| Component         | CAS No   | Ozone Depletion Potential |
|-------------------|----------|---------------------------|
| Isopropoxyethanol | 109-59-1 | Not listed                |

### Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

## SECTION 14: TRANSPORT INFORMATION

### Road and Rail Transport

|                         |                          |
|-------------------------|--------------------------|
| UN-No                   | UN1993                   |
| Proper Shipping Name    | FLAMMABLE LIQUID, N.O.S. |
| Technical Shipping Name | 2-Isopropoxyethanol      |
| Hazard Class            | 3                        |
| Packing Group           | III                      |

### IATA

|                         |                          |
|-------------------------|--------------------------|
| UN-No                   | UN1993                   |
| Proper Shipping Name    | FLAMMABLE LIQUID, N.O.S. |
| Technical Shipping Name | 2-Isopropoxyethanol      |
| Hazard Class            | 3                        |
| Packing Group           | III                      |



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## IMDG/IMO

UN-No UN1993  
 Proper Shipping Name FLAMMABLE LIQUID, N.O.S.  
 Technical Shipping Name 2-Isopropoxyethanol  
 Hazard Class 3  
 Packing Group III  
 Marine Pollutant No hazards identified

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

#### International Inventories

| Component         | CAS No   | KECL     | TSCA | EINECS    | IECSC | DSL | NDSL | PICCS | ENCS | ISHL | AICS |
|-------------------|----------|----------|------|-----------|-------|-----|------|-------|------|------|------|
| Isopropoxyethanol | 109-59-1 | KE-23948 | X    | 203-685-6 | X     | X   | -    | X     | X    | X    | X    |

| Component         | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Isopropoxyethanol | 109-59-1 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

| Component         | CAS No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential |
|-------------------|----------|----------------|------------------------------|---------------------------|
| Isopropoxyethanol | 109-59-1 | Not applicable | Not applicable               | Not applicable            |

#### Korean National Regulations

| Component         | CAS No   | Act on Registration and Evaluation of Chemical Substances (K-REACH) | Authorised Chemicals | Existing Substances Subject to Registration |
|-------------------|----------|---------------------------------------------------------------------|----------------------|---------------------------------------------|
| Isopropoxyethanol | 109-59-1 | Annex 1 - KE-23948                                                  | Not applicable       | Not applicable                              |

| Component         | CAS No   | Chemical Control Act - Toxic Chemicals | Chemical Control Act - Prohibited Chemicals | Chemical Control Act - Use Restricted Chemicals |
|-------------------|----------|----------------------------------------|---------------------------------------------|-------------------------------------------------|
| Isopropoxyethanol | 109-59-1 | Not applicable                         | Not applicable                              | Not applicable                                  |

| Component         | CAS No   | Chemical Control Act - Accident Precaution Chemicals (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures) |
|-------------------|----------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Isopropoxyethanol | 109-59-1 | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |

| Component         | CAS No   | Waste Control Law | Ministry of Environment - CMR risk | Ministry of Environment - Critically Controlled Substance |
|-------------------|----------|-------------------|------------------------------------|-----------------------------------------------------------|
| Isopropoxyethanol | 109-59-1 | Not applicable    | Not applicable                     | Not applicable                                            |

| Component | CAS No | ISHA - Harmful Agents Subject to Work | ISHA - Prohibited substances | ISHA - Substances requiring permission |
|-----------|--------|---------------------------------------|------------------------------|----------------------------------------|
|-----------|--------|---------------------------------------|------------------------------|----------------------------------------|

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|                   |          | Environment Monitoring |                |                |
|-------------------|----------|------------------------|----------------|----------------|
| Isopropoxyethanol | 109-59-1 | Not applicable         | Not applicable | Not applicable |

| Component         | CAS No   | ISHA - Substances subject to control | ISHA - Harmful Agents Requiring Health Examination | ISHA - Permissible Exposure Limits |
|-------------------|----------|--------------------------------------|----------------------------------------------------|------------------------------------|
| Isopropoxyethanol | 109-59-1 | Not applicable                       | Not applicable                                     | Not applicable                     |

| Component         | CAS No   | ISHA - Subject to Process Safety Reports (minimum quantity) | ISHA - Threshold Limit Values (TLVs) Chemicals | ISHA - Special management materials |
|-------------------|----------|-------------------------------------------------------------|------------------------------------------------|-------------------------------------|
| Isopropoxyethanol | 109-59-1 | 5000 kg                                                     | TWA: 25 ppm<br>Skin                            | Not applicable                      |

**National Fire Association - Dangerous Substances** Minimum quantity requiring a permit

| Component         | CAS No   | Class 1 - Oxidising solids | Class 2 - Flammable solid | Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet | Class 4 - Flammable liquids             | Class 5 - Self-reactive substances | Class 6 - Oxidising liquids |
|-------------------|----------|----------------------------|---------------------------|----------------------------------------------------------------------------------|-----------------------------------------|------------------------------------|-----------------------------|
| Isopropoxyethanol | 109-59-1 | Not applicable             | Not applicable            | Not applicable                                                                   | 4. Group 2 Petroleum (Insoluble) 1000 L | Not applicable                     | Not applicable              |

## Control Parameters

| Component         | CAS No   | Korea               | ACGIH - Biological Exposure Indices |
|-------------------|----------|---------------------|-------------------------------------|
| Isopropoxyethanol | 109-59-1 | TWA: 25 ppm<br>Skin | Not listed                          |

## US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

| Component         | CAS No   | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|-------------------|----------|----------------------------------|----------------------------|
| Isopropoxyethanol | 109-59-1 | Not applicable                   | Not applicable             |

**CERCLA** Not applicable

| Component         | CAS No   | CERCLA Extremely Hazardous Substances RQs | Hazardous Substances RQs | SARA 313 - Threshold Values % |
|-------------------|----------|-------------------------------------------|--------------------------|-------------------------------|
| Isopropoxyethanol | 109-59-1 | Not applicable                            | Not applicable           | 1.0 %                         |

## GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Warning.

H226 - Flammable liquid and vapor. H319 - Causes serious eye irritation. H312 + H332 - Harmful in contact with skin or if inhaled.

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water. P280 - Wear protective gloves/protective clothing/eye protection/face protection. P310 - Immediately call a POISON CENTER or doctor/physician. P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

## SECTION 16: OTHER INFORMATION

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## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

## Prepared By

Health, Safety and Environmental Department

## Creation Date

15-Apr-2015

## Revision Date

09-Jun-2024

## Revision Number

6

## Revision Summary

New emergency telephone response service provider.

## MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**